## Camera specifications

**Samsung Galaxy S23:**
- 50 MP Samsung S5KGN3 primary camera, OIS and dual-pixel PDAF, f/1.8 aperture
- 10 MP telephoto camera, 3x optical zoom and OIS
- 12 MP Sony IMX564 ultrawide shooter
- 12 MP Samsung S5K3LU selfie camera, dual-pixel PDAF
- Video at up to 8K@30fps

**Google Pixel 7:**
- 50 MP Samsung ISOCELL GN1 primary camera, OIS and PDAF, f/1.9 aperture
- 12 MP Sony IMX381 ultrawide sensor
- 10.8 MP Selfie Camera
- Video at up to 4k@60fps

**Apple iPhone 14:**
- 12 MP Primary camera, dual pixel PDAF and OIS, f/1.5 aperture
- 12 MP ultrawide lens
- 12 MP Selfie Camera, PDAF
- Video at up to 4K@60fps

shadow areas, we found the iPhone 14 to be slightly lacking in details compared to the other two.

In tricky lighting, the Samsung Galaxy S23 shows its prowess by exposing highlights and shadows accurately. In the pictures attached below, you can see that the Galaxy S23 has controlled the highlights well while also maintaining detail in the shadows of the tree bark. The Pixel 7 also does a great job, however, there's a bit of detail in the shadows being crushed in tricky lighting. The iPhone 14 has an issue of occasional lens flare in such situations. In most photos though, the dynamic range on all three phones is on point but Samsung's colours are a bit too saturated which some may like and some may despise. The images coming from the S23 are vibrant and look social-media-ready, but occasionally this can appear slightly unnatural. The Pixel 7 definitely has the most true-to-life colours, especially for human skin tones. The S23 does have a tendency to whitewash brown human subjects at times, while the Pixel and iPhone try to keep a more natural look. All three phones also have sizable lenses, so the natural bokeh effect is quite good on all of them with minimal fringing, however, the Pixel 7's bokeh effect looks the most natural.

As for detail retention, we have very little to complain about. In most instances you are going to get pristine shots from all three phones' primary cameras with tonnes of detail packed in the highlight and shadow areas.

iPhone 14 (left), Pixel 7 (middle), Galaxy S23 (right)

iPhone 14 (left), Pixel 7 (middle), Galaxy S23 (right)

## Portrait

Moving on to portrait photos, by default, the Google Pixel 7's 1x portrait shot in cropped in compared to the iPhone and Samsung phones. Pixel's default portrait images look more like 1.5x rather than 1x. Again, the Pixel 7 and the iPhone 14 have a more natural approach to handling skin tones, with the Pixel edging out in colour accuracy by a bit. Samsung can distort the colour of human subjects, making their skin appear more pale or cool-toned that it is in reality.

However, when it comes to edge detection, Samsung knocked it out of the park. It has easier time getting a clean cutout in most instances with minimal error. Even the area between the arms and the waist, that often gets missed by some sensors, is blurred out by the Samsung Galaxy S23's camera. However, since this is done via AI, there are times when all three phones miss out on getting the perfect cutout.

Another advantage on the Galaxy S23 is the presence of a 3x optical zoom telephoto lens. The telephoto camera